

Numerical investigation of torrefied empty fruit bunch co-firing in a 500 MWe pulverized coal boiler: turbulence, particle and NO_x modelling

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ABSTRACT

Co-firing biomass with coal is a practical route to reduce fossil carbon emissions from existing utility boilers. Empty fruit bunch (EFB), a residue of palm oil production, is abundant but its high moisture, alkali and volatile content complicate direct combustion. Torrefaction at 250–300 °C improves grindability and energy density. In this sample study, a three-dimensional computational fluid dynamics (CFD) model of a 500 MWe tangentially fired boiler is used to examine co-firing ratios of 0, 10, 20 and 30% on a thermal basis. A realizable k- ϵ turbulence model was selected to calculate the effect of turbulence in the boiler. Particles were tracked with the discrete phase model (DPM), radiation was solved with the discrete ordinates (DO) model, and NO_x formation was evaluated as a post-process. The illustrative results show that the wall heat flux in the burner zone decreases with increasing co-firing ratio, while the furnace exit gas temperature rises slightly. NO_x at the furnace exit decreases as the co-firing ratio increases, which is attributed to lower fuel nitrogen and stronger volatile-driven reducing zones.

Keywords: EFB; torrefaction; coal co-firing; realizable k- ϵ ; DPM; NO_x; heat flux; boiler efficiency

1. Introduction

Utility boilers designed for bituminous coal represent a large installed capacity that will remain in service for decades. Retrofitting these units for partial biomass firing allows operators to reduce net carbon emissions without replacing the steam cycle. The main technical questions concern flame stability, slagging and fouling, burnout, and the redistribution of heat absorption between the furnace and the convective pass.

Raw EFB is fibrous and difficult to pulverize in existing coal mills. Torrefaction, a mild pyrolysis in an inert atmosphere, removes moisture and part of the hemicellulose, producing a brittle, hydrophobic solid. The resulting torrefied EFB has a higher heating value and can be co-milled with coal at moderate ratios. However, its volatile matter remains considerably higher than that of coal, which changes ignition behaviour near the burners.

Computational models are widely used to anticipate these effects before plant trials. The reliability of such predictions depends on the choice of turbulence closure, the devolatilization and char combustion submodels, and the treatment of radiation in a particle-laden flue gas. This sample manuscript deliberately uses a conventional model set so that each choice can be inspected in the reader.

The objectives of this illustrative study are: (i) to quantify how the co-firing ratio changes the temperature and heat flux distribution, (ii) to evaluate the change in NO_x at the furnace exit, and (iii) to discuss the implications for boiler efficiency and operation.

2. Numerical method

2.1 Boiler geometry and operating conditions

The modelled furnace is a tangentially fired, single-reheat boiler with four burner elevations on each corner. The furnace cross section is 15.6 m × 15.6 m and the height from the hopper to the furnace exit plane is 58 m. Separated over-fire air (SOFA) ports are installed above the top burner elevation. The computational domain extends from the hopper to the exit of the platen superheater region.

Table 1. Proximate and ultimate analysis of the fuels (illustrative, as received).

Property	Coal	Torrefied EFB
Moisture (wt%)	12.4	3.1
Volatile matter (wt%)	31.2	58.7
Fixed carbon (wt%)	46.9	29.4
Ash (wt%)	9.5	8.8
Nitrogen, dry (wt%)	1.42	0.61
HHV (MJ/kg)	25.8	21.3

2.2 Turbulence model

A realizable k - ϵ turbulence model was selected to calculate the effect of turbulence in the boiler. Compared with the standard k - ϵ model, the realizable formulation satisfies mathematical constraints on the Reynolds stresses and provides improved predictions for swirling and rotating flows, which are characteristic of tangentially fired furnaces. Standard wall functions were applied at the water walls.

The transport equation for the turbulent kinetic energy k is written as

$$\partial(\rho k)/\partial t + \partial(\rho k u_j)/\partial x_j = \partial/\partial x_j[(\mu + \mu_t/\sigma_k) \partial k/\partial x_j] + G_k - \rho\epsilon \quad (1)$$

and the turbulent viscosity is obtained from $\mu_t = \rho C_\mu k^2/\epsilon$, where C_μ is not constant but a function of the mean strain and rotation rates. The realizable model therefore avoids non-physical negative normal stresses in regions of strong streamline curvature.

2.3 Discrete phase model

Particle trajectories of coal and torrefied EFB were tracked with the discrete phase model (DPM) in a Lagrangian frame. The particle size distribution of each fuel was represented by a Rosin–Rammler function with mean diameters of 65 μm for coal and 210 μm for torrefied EFB. Turbulent dispersion was included with a stochastic tracking approach. Devolatilization was described by a single kinetic rate model of the Arrhenius form

$$k_v = A_v \exp(-E_v / RT_p) \quad (2)$$

with separate kinetic parameters for each fuel. Char combustion was modelled with a kinetics/diffusion-limited rate. Two-way coupling between the particles and the gas phase was applied for mass, momentum and energy.

2.4 Radiation and gas-phase combustion

Radiative heat transfer was solved using the discrete ordinates (DO) model. The absorption coefficient of the gas mixture was computed with the weighted-sum-of-gray-gases model, and particle radiation was included through an effective particle emissivity. Volatile combustion was represented by a two-step global mechanism with the eddy-dissipation model.

2.5 NO_x post-processing

NO_x formation was evaluated as a post-process on the converged flow field. Thermal NO_x was calculated with the extended Zeldovich mechanism, and fuel NO_x was calculated assuming that fuel nitrogen is released with the volatiles as HCN and with the char as NO. NO reburning by hydrocarbon radicals was included. Because the post-processing does not

feed back into the flame, the predicted NO_x should be interpreted as a relative trend between cases.

2.6 Cases and boundary conditions

Four cases were simulated: 0% (coal only), 10%, 20% and 30% torrefied EFB on a thermal basis. The total fuel heat input and the overall excess air ratio (1.18) were held constant. The primary air flow was adjusted to maintain the mill outlet air-to-fuel ratio, while the secondary and SOFA flows were scaled proportionally.

3. Results and discussion

3.1 Temperature distribution

For the coal-only case the peak gas temperature of approximately 1,720 K occurs in the central swirl region between the second and third burner elevations. With increasing co-firing ratio the flame front moves closer to the burner outlets because the volatiles of torrefied EFB ignite earlier. The peak temperature decreases slightly, to approximately 1,680 K for the 30% case, owing to the lower adiabatic flame temperature of the blend.

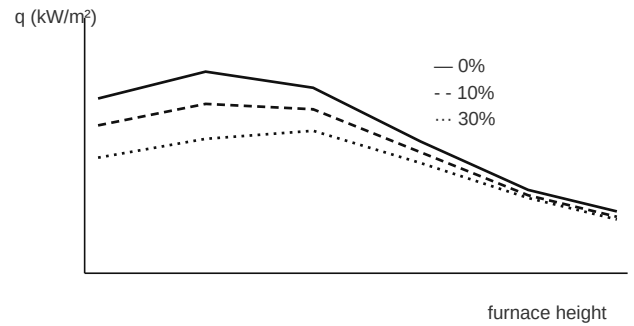


Fig. 1. Illustrative wall heat flux profiles along the furnace height for three co-firing ratios (synthetic data).

3.2 Wall heat flux

The heat flux to the furnace wall in the burner zone decreased as the co-firing ratio increased. At the 30% co-firing ratio the area-averaged heat flux in the burner zone was 7.8% lower than in the coal-only case. Above the SOFA elevation the differences become small, and at the furnace exit the gas temperature is 18 K higher for the 30% case. This shift of heat absorption from the radiant furnace towards the convective pass is consistent with the lower particle emissivity and the larger particle size of torrefied EFB.

A lower radiant heat absorption implies that superheater and reheater spray flows may need to increase to control steam temperature. Operators should therefore consider the steam-side consequences of the redistributed heat flux, not only the combustion performance.

3.3 Particle burnout

Despite the larger mean diameter, the torrefied EFB particles reach high burnout because of their high volatile content and porous char. The unburned carbon in fly ash at the furnace exit increases from 2.1% (coal only) to 2.6% at the 30% co-firing ratio, indicating that milling fineness is a sensitive parameter for higher ratios.

Table 2. Illustrative furnace exit results by co-firing ratio.

Co-firing ratio	FEGT (K)	NOx (mg/Nm ³ , 6% O ₂)	UBC (%)
0%	1,402	312	2.1
10%	1,407	291	2.2
20%	1,413	268	2.4
30%	1,420	247	2.6

3.4 NOx emissions

NOx at the furnace exit decreases as the co-firing ratio increases. Two mechanisms contribute: the lower nitrogen content of torrefied EFB reduces the fuel nitrogen input, and the early release of volatiles creates fuel-rich zones near the burners in which NO is reduced. The thermal NOx contribution changes little because the peak temperatures differ by less than 50 K between cases.

It should be noted that the NOx post-processing approach neglects the influence of alkali species and of detailed nitrogen partitioning in the biomass. The predicted reduction of 21% at the 30% co-firing ratio should therefore be verified against measurements.

3.5 Implications for boiler efficiency

Boiler efficiency is affected by the flue gas exit temperature, unburned carbon and moisture losses. The higher furnace exit gas temperature and slightly higher unburned carbon both reduce efficiency, whereas the lower fuel moisture of torrefied

EFB partly compensates. In this illustrative energy balance the net change in boiler efficiency is within ± 0.3 percentage points for co-firing ratios up to 20%.

4. Limitations

The steady RANS approach cannot resolve the unsteady flame dynamics near the burners. The single kinetic rate devolatilization model and the global gas-phase mechanism limit the accuracy of local temperature and species predictions. Ash deposition, which is a major concern for alkali-rich biomass, was not modelled and may change the wall heat flux over time.

5. Conclusions

A CFD model of a 500 MWe tangentially fired boiler was used to examine torrefied EFB co-firing up to 30% on a thermal basis. Within the assumptions of this sample study: (1) the burner zone heat flux decreases and the furnace exit gas temperature increases with the co-firing ratio; (2) NOx at the furnace exit decreases as the co-firing ratio increases; and (3) the net effect on boiler efficiency is small up to 20%. Plant measurements are required before these trends are used for design decisions.

Nomenclature

k turbulent kinetic energy (m²/s²); ϵ dissipation rate (m²/s³); μ_t turbulent viscosity (Pa·s); G_k production of k ; A_v pre-exponential factor (1/s); E_v activation energy (J/mol); T_p particle temperature (K); FEGT furnace exit gas temperature; UBC unburned carbon in fly ash.

References

References are intentionally omitted from this synthetic sample so that no citation is fabricated.